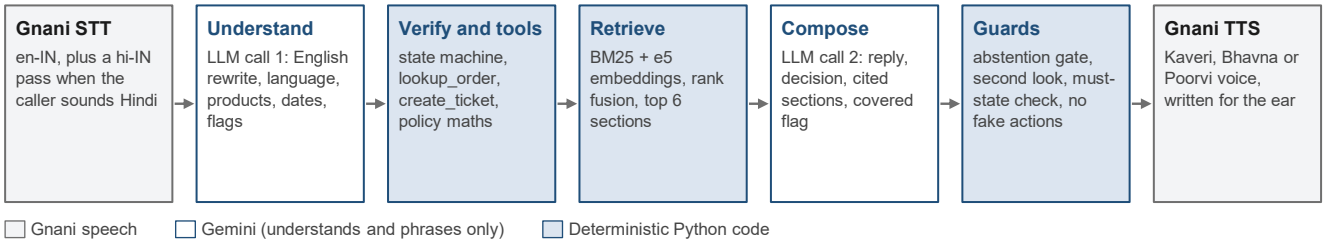
VoiceOps 2026: ZentroMart voice support agent

Capstone report | Team Dstonks | 25 October 2026

Author	Shreshtha (Team Dstonks)	Repository	github.com/nightraven4545/voiceops-2026
Language model	Gemini gemini-3.1-flash-lite, free tier, OpenAI-compatible API	Speech	Gnani gnani-vachana 0.8.2: Prisma STT, Timbre v2.5 TTS
Code	run.py, src/ (4 modules), eval/, tests/	Results	Full agent 96% fully correct, conversations 98% of turns, tool calls 100% exact

Figure 1: One voice turn, from caller audio to spoken reply



Source: src/ and run.py. The LLM never sees phone, pincode or customer ID, and never decides an escalation.

1 What we built, step by step

Step 1: prompt only

One LLM call. The system prompt sets the role (ZentroMart's AI support assistant), the limits (no promised refunds or actions, say when unsure) and the length. No documents.

Step 2: grounded in the documents

Table 1: Retrieval and answering design

Component	Choice
Chunking	Each ## [SECTION-ID] heading is one chunk: 84 chunks of 6–135 words, so a rule is never cut off from its heading
Metadata	Every chunk keeps its doc_type, status and effective_from
Retrieval	BM25 (bm25s, with a romanized Hindi stoplist) and multilingual embeddings (intfloat/multilingual-e5-small, numpy index), fused by reciprocal rank
Query rewrite	Before retrieval, an LLM call rewrites the message into a standalone English query. The raw message also counts, at half weight
Answering	The model sees the top 6 sections (12 on a second look) sorted and labelled by authority: current policy, agent guide, help, FAQ, marketing, with superseded sections marked. It must return the section IDs it used

Source: src/ (retrieval module) and the dataset knowledge base.

Step 3: a real conversation

Code owns everything that must not go wrong. The model only understands the message and phrases the reply.

- **Verification state machine.** An order ID triggers lookup_order(order_id) and a request for the phone's last 4 digits or the delivery pincode. The answer triggers the verified lookup. One retry per order, then refusal (IDV-3). Naming another order does not reset it.
- **Private data never reaches the LLM.** Phone, pincode and customer ID are stripped from tool results.

Table 2: Policy maths in Python (src/rules.py, unit tested against the practice gold)

Rule	How code applies it
Refund business days	Count from the day after initiation, Monday to Saturday, skipping the holiday table. This reproduces all 9 dated refund answers
Return windows	Calendar days, with the RET-3 exception (pre-August electronics: 10 days) and RET-8 (festive fashion: 30 days)
Refund arrival	Ranges per payment method, and the "more than 3 business days overdue" check
Fees and references	The ₹49 pickup fee, refund timing after refusing at the door, and the refund reference number
Pincodes and cities	Delivery, cash on delivery, Express and metro status for a pincode or city the caller names, from the pincode table (DEL-5, DEL-1)

Source: src/rules.py and tests/.

Policy-owned decisions

- The ESC-1 triggers force an escalation: human requested, fraud, legal, safety, two or more earlier contacts, item above ₹10,000, or an overdue refund. The model cannot escalate on its own.
- A compensation demand is refused unless it is the one case DEL-4 pays for (a late Express order). The brief's example also raises a ticket here. The practice gold refuses without one, and we follow the gold. A caller who then asks for a person still gets a ticket and a handover.

Table 3: Guards applied to every reply

Guard	Behaviour
Ticket honesty	A ticket is only mentioned after create_ticket returns created=True
No fake actions	Replies claiming a cancellation, approval or credit, or showing private digits, are regenerated
Evidence check	Evidence IDs that were not shown to the model are dropped
Abstention gate	The model must mark whether the evidence actually answers the question. If not, code replaces the reply with "I don't have that information" plus an offer of a human colleague
Second look	Before giving up, the agent composes once more with the top 12 sections. The new reply is used only if it is a plain answer, covered, cites a newly added content section (not a guide note, not every section) and passes the other guards. Otherwise it abstains as before
Must-state post-check	Where a policy requires a computed fact (the refund reference number when a verified customer asks about a refund, REF-6, or the 7–10 day refund timing when a shipped order's cancellation is refused, CAN-2), code checks that the reply contains its numbers. If not, one rewrite is requested. It is kept only if it states the fact, keeps every number of the first reply and passes the other guards. On any failure the first reply stands

Source: src/ (agent module).

Step 4: voice

Table 4: Voice handling in run.py

Stage	Behaviour
Input	Any audio format is converted to 16 kHz mono, split into pieces of at most 30 s and transcribed with Gnani en-IN
Hindi second pass	If the transcript looks Hindi (Hindi function words or Hindi number words), a second hi-IN pass runs and the agent receives both transcripts. en-IN writes spoken Hindi numbers as words ("saat hazaar…"), hi-IN returns digits
Speech output	Replies are written for the ear (dates with month and year, order IDs in groups of four) and spoken with Kaveri (en-IN), Bhavna (hi-IN) or Poorvi (Hinglish, via the documented hi-en REST code, which the SDK rejects)
Robustness	Silence gets "Sorry, I didn't catch that". Sub-folders of audio are treated as one conversation. A failing file never stops the batch

Source: run.py and src/ (speech module).

2 Evaluation

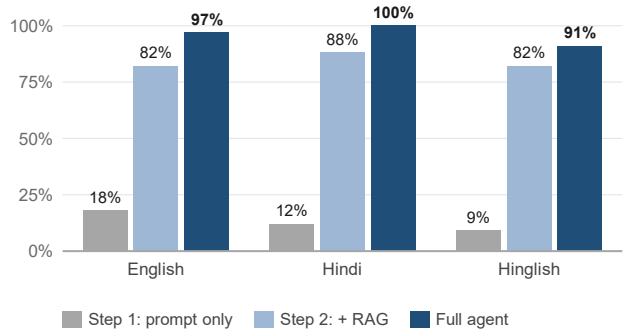
All numbers are on the dataset's practice files and come from eval/evaluate.py. A reply is **fully correct** only if the decision is one of the acceptable ones, every must_include fact is present (dates matched by value across English and Hindi formats, numbers as whole numbers) and no must_not_include phrase appears. Gold-evidence recall is reported separately. Result files are in eval/results/<model>/.

Table 5: Step 1 vs Step 2 vs full agent: 52 single questions, gemini-3.1-flash-lite

Measure	Step 1: prompt only	Step 2: + RAG	Full agent	Full agent, noisy asr_text
Fully correct	15%	83%	**96%**	**96%**
Required facts present	23%	87%	98%	98%
Acceptable decision	75%	90%	98%	98%
Forbidden phrase used	0%	0%	0%	0%
Gold-evidence recall	0.00	0.72	0.83	0.85
Fully correct: English / Hindi / Hinglish	18 / 12 / 9%	82 / 88 / 82%	97 / 100 / 91%	97 / 88 / 100%

Source: eval/results/gemini-3.1-flash-lite/, modes baseline, rag, full and full --asr.

Figure 2: Fully correct by language



Source: eval/results/gemini-3.1-flash-lite/, clean text.

Table 6: Retrieval of gold section IDs, 46 answerable questions

Query used	R@1	R@3	R@5	MRR	R@5 Hindi
Raw customer text	0.30	0.48	0.68	0.50	0.43
English rewrite + raw at half weight	0.43	0.78	**0.90**	0.69	**0.86**
Same, on noisy asr_text	0.45	0.74	0.85	0.69	–

Source: eval/evaluate.py --mode retrieval.

The rewrite lifts Recall@5 from 0.68 to 0.90, and from 0.43 to 0.86 on Hindi questions.

What the jumps show

- **Step 1 to Step 2:** retrieval is worth 68 points. Without documents the model answers fluently and wrongly, for example the generic "5–7 days" for every refund.
- **Step 2 to full agent:** worth another 13 points, from catalog rows (warranty months, the Assured rule for marketplace sellers), the date calculators, escalation decided by policy rules, and the abstention gate.
- **Noisy transcripts:** speech recognition text (asr_text: spelled out digits, misheard words) costs nothing overall, thanks to the transcript normaliser and the English rewrite.

Conversations: 23 conversations, 96 customer turns

- 98% of turns fully correct, **100% correct decisions, 100% exactly the expected tool calls** (lookup_order with and without the verification value, create_ticket), evidence recall 0.94. Per scenario, 100% everywhere except compensation demand and refund status (88% each).
- By split: dev 74/75, held-out test 20/21. The must-state post-check and the second look were both built from dev failures, and no test turn triggers either, so test is unchanged (20/21 before and after). It shows no regression, not a held-out gain. On single questions the second look ran on the 8 correct abstentions (dev, plus 1 test on ASR text) and all 8 still abstained.

Voice end to end

- A 17-file test set, made with Gnani TTS plus a silent clip, ran through run.py with the batch never stopping: 8 questions in three languages, two 4-turn conversations (Hinglish high-value damage, Hindi failed verification) and silence.
- In the Hindi conversation the caller's spoken code was read correctly: the hi-IN pass gives "1,266", which the extractor turns into 1266. The agent then made the gold tool calls: verify, ask again, refuse.
- **Latency.** On the first live run a voice turn took 7–50 s on the free tier, much of it rate-limit waiting and retries. A rerun with the LLM answers cached, so speech only, took a median of 6.5 s (3–20 s): the Gnani STT and TTS share of a turn. A second look or must-state rewrite adds one LLM call (typically 2–4 s) on the few turns that need it. These optional calls skip client retries, so a rate limit makes them fall back at once instead of waiting.

Table 7: Model comparison, same code (gemini-3.5-flash-lite was used first)

Measure	gemini-3.1-flash-lite (submission)	gemini-3.5-flash-lite (earlier)
Fully correct, Step 1	15%	21%
Fully correct, Step 2	83%	67% (before the prompt fixes)
Fully correct, full agent	**96%**	92%
Conversations, turns fully correct	98%	Cut short by the daily quota
Time per LLM call in our probe	2.3 s	5.2 s

Source: eval/results/<model>/.

3 What went wrong, and what we fixed

Table 8: Problems the evaluation found, and the fixes

Problem the evaluation found	Fix
The section parser captured one section per file (.* under DOTALL swallowed the rest)	Anchored the heading pattern. A test asserts 84 sections
Dates in the query ("17 October 2026") pulled in sections that merely mention October	Stripping dates helped English but hurt festive-sale questions, where the date is the topic, so it was reverted. The English rewrite solved it
The model invented answers to unanswerable questions ("GST invoice mil sakta hai")	covered flag and abstention gate
The model escalated on its own, and treated an Express-late question as compensation	Escalation comes only from the ESC-1 rules. A DEL-4 rule for compensation
Follow-ups ("and what if I'm not at home?"): the raw follow-up outvoted the context-aware rewrite	The raw query counts at half weight
Spoken Hindi numbers: en-IN gave "saat hazaar…", and Gnani's hi-IN formatting wrote "7,746 1,439"	Hindi number words trigger the hi-IN pass. The extractor drops thousands separators and joins split digit groups
A Hindi caller reading out digits was answered in Hinglish	The conversation keeps its language on such turns
In 3 conversation turns the decision was right but a computed fact was left out (refund timing after refusing a parcel, the refund reference number)	Must-state post-check: a rewrite only for those turns (3 extra LLM calls in the whole evaluation). Conversations 93% to 96%
"And what if I'm not at home?" got "I don't have that information": the answer (DEL-3, 3 delivery attempts) ranked 8th and 11th, outside the top 6	Second look with 12 sections before abstaining, accepted only on new cited content. 10 extra LLM calls across all evaluations. Conversations 96% to 98%
Our own review found that naming a different order reset the verification retry, allowing unlimited guesses	Failed attempts are counted per order for the whole conversation
Gnani's reply WAVs carry placeholder sizes (0xFFFFFFFF), so strict readers saw about 24 h of audio	The file is rewritten once with real sizes after synthesis
The free LLM tier ran out mid-evaluation (about 550 calls a day per model)	On-disk response cache. Failed rows refilled on a later run
Tried down-weighting superseded sections in retrieval	Rejected: it dropped the gold RETv3-2 for the pre-August electronics case the policy still covers

Source: eval/results/ before and after each change, PROJECT_STATE.md checklist history.

Still open, and what we would fix next

- **Two single-question failures.** One unanswerable question (store pickup) still got an invented answer the gate did not catch, and one asked an unnecessary "which colour?" before answering.
- **Correct but differently worded.** 2 failures use a Hindi synonym the strict string scorer does not list ("Mujhe khed hai" for "sorry", "क्वालिटी चेक" for "quality check"). A semantic scorer would pass them.
- **Spoken SKUs.** "ELE zero zero one two" comes back from STT as "s q le 12" and cannot be recovered. Gnani's word boosting (bias_list, REST only) is the next step.

4 Bonus work

- **Tests:** 80 automated tests, including scripted-LLM conversation tests (no API calls) for verification, ticket rules, abstention and its second look, the must-state post-check and language.
- **Evaluation tooling:** a Gnani TTS test-audio generator, results stored per model, retrieval scored with and without the rewrite.
- **Live demo:** a Gradio app (app.py) showing each turn's decision, cited sections, tool calls and verification state.